

- 9 . Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection.

OUTPUT :

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("source.txt", O_RDONLY);

    if (source == -1)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination == -1)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead)
        {
            perror("Error writing file");
            close(source);
            close(destination);
            return 1;
        }
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

```
bhargav08@Bhargava:~/OS$ nano practical9A.c
bhargav08@Bhargava:~/OS$ gcc practical9A.c -o practical9A
bhargav08@Bhargava:~/OS$ ./practical9A
File copied successfully.
```

9(B):

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[20];

    fd = open("source.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    lseek(fd, 0, SEEK_SET);

    read(fd, buffer, sizeof(buffer) - 1);

    buffer[sizeof(buffer) - 1] = '\0';

    printf("Data: %s\n", buffer);

    close(fd);

    return 0;
}
```

```
bhargav08@Bhargava:~/OS$ nano practical9B.c
bhargav08@Bhargava:~/OS$ gcc practical9B.c -o practical9B
bhargav08@Bhargava:~/OS$ ./practical9B
Data: Hello from source
```

9(C) :

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input, output;

    input = open("input.txt", O_RDONLY);

    if (input == -1)
    {
        perror("Error opening input file");
        return 1;
    }

    output = open("output.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (output == -1)
    {
        perror("Error opening output file");
        close(input);
        return 1;
    }

    // Redirect standard input
    dup2(input, STDIN_FILENO);

    // Redirect standard output
    dup2(output, STDOUT_FILENO);

    close(input);
    close(output);

    char buffer[1024];
    ssize_t n;

    while ((n = read(STDIN_FILENO, buffer, sizeof(buffer))) > 0)
    {
        write(STDOUT_FILENO, buffer, n);
    }

    return 0;
}
```

```
bhargav08@Bhargava:~/OS$ nano practical9C.c
bhargav08@Bhargava:~/OS$ gcc practical9C.c -o practical9C
bhargav08@Bhargava:~/OS$ ./practical9C
bhargav08@Bhargava:~/OS$ nano input.txt
bhargav08@Bhargava:~/OS$ nano output.txt
hfcuyabfsvbighsbvikhgsiuviusvnjkhgsjkhs
```